

Models

Computational Neuroscience

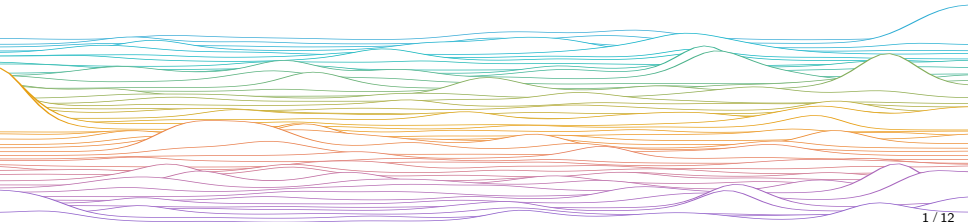
University of Bristol

Slides adapted from C O'Donnel

M Rule

Learning outcomes:

- ▶ Become inspired to creatively imagine models that are entirely phenomenological, provably wrong from the outset, — yet interesting, and maybe even useful.
- ▶ Be aware of some models we will cover in this course



What is a model?

Word models

Describe system's components and their interactions

Describe connections, causality, effect directions, magnitudes, & timescales

Often all you need!

Math/computational models: Tools for

Scientific Enquiry

- ▶ A logical, specific, and falsifiable type of word model
- ▶ Computer code ~ executable mathematics

Data visualization, inference

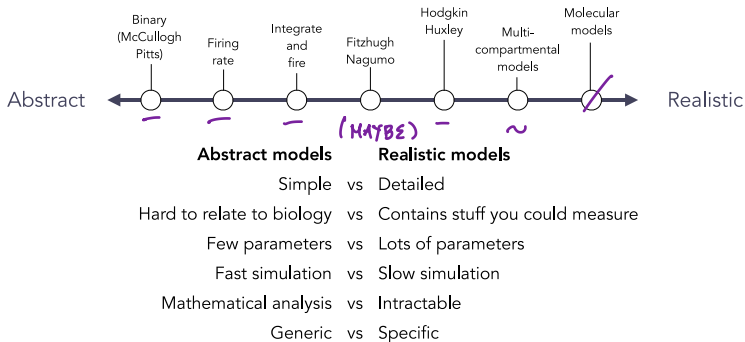
- ▶ A compact summary of experimental results

Engineering (neurotechnology, medicine)

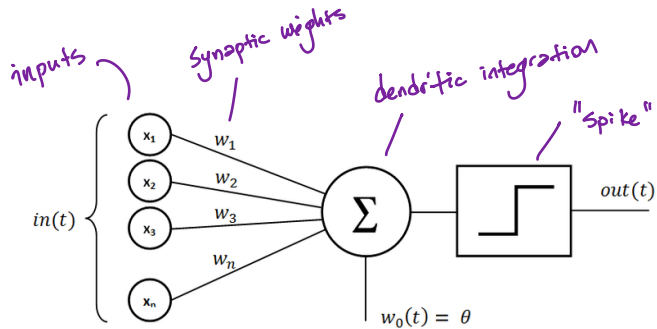
- ▶ Accurate, quantitative modelling of collective dynamics in epilepsy used to design closed-loop neurotechnology devices

All models are, in some way, phenomenological

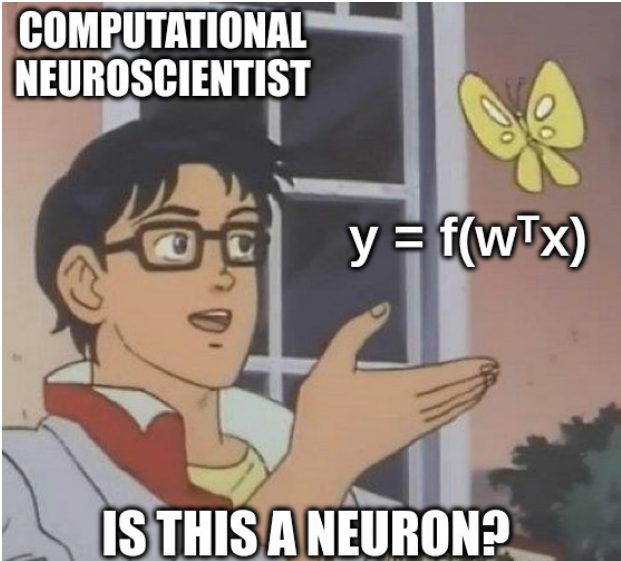
Models of Single Neurons



C. O'Donnel (2022)



**COMPUTATIONAL
NEUROSCIENTIST**


$$y = f(\mathbf{w}^T \mathbf{x})$$

IS THIS A NEURON?

Vector notation
of diagram
on previous
slide

Which model is best for my problem?

“It can scarcely be denied that the supreme goal of all theory is to make the irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience”

— Einstein (1933) *Lecture at the Royal Albert Hall*

Choose the model that best matches the granularity of your scientific question

Choice dictated by:

- ▶ Available data
- ▶ Phenomena of interest
- ▶ Computational practicalities
- ▶ ~~Historical precedent~~

“Am I allowed to do that?”

Usually, yes

Modelling Scales

... Quantum Chemistry, Molecular dynamics

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Molecules
Gillespie, Master Equation
Concentrations
Mass-Action Kinetics
Conductance Models
Hodgkin–Huxley
Spiking Models
Leaky Integrate and Fire
Rate Neurons
Neural Mass/Field Models
Poisson Neurons
Generalized Linear Models
Binary Neurons
McCulloch–Pitts, Hopfield, Perceptron

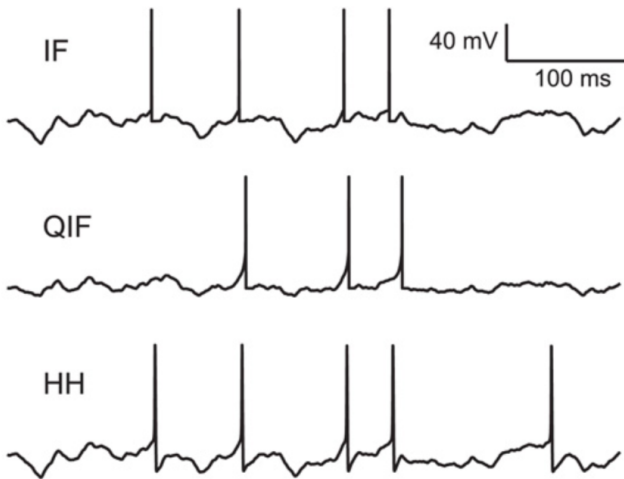


Computational
Efficiency,
Mathematical
Tractability

Phenomenological,
Qualitative

Cognitive Neuroscience, Psychology ...

How frustrated you would be if you went to enormous lengths to robustly approximate a LIF neuron using messy ionic conductances, only to turn around and discover that some scientist had declared this model “not biologically plausible”?



Voltage from three different model neurons in response to the same identical input current; Brette (2015) PLoS Comput Biol

end